

### Question 1

Link delay rate = 8 Gbps = 1 GB/s

OCN:

Propagation delay =  $0.005\text{m} / 200000000\text{m/s} = 2.5 \times 10^{-11}\text{s} = 0.025\text{ns}$

Transmission delay =  $100\text{bytes} / 1000000000\text{bytes/s} = 1 \times 10^{-7}\text{s} = 100\text{ns}$

Processing delay =  $0.5\text{ns} \times 100 \text{ bytes} = 50\text{ns}$

Total delay =  $150.025\text{ns}$

Percentage that is propagation delay =  $0.025\text{ns}/150.025\text{ns} = 0.000166 = 0.0166\%$

SAN:

Propagation delay =  $5\text{m} / 200000000\text{m/s} = 25\text{ns}$

Transmission delay =  $100\text{ns}$

Processing delay =  $0.5\text{ns} \times 100 \text{ bytes} + 300\text{ns} = 350\text{ns}$

Total delay =  $475\text{ns}$

Percentage that is propagation delay =  $25\text{ns}/475\text{ns} = 0.052632 = 5.2632\%$

LAN:

Propagation delay =  $5000\text{m} / 200000000\text{m/s} = 25000\text{ns}$

Transmission delay =  $100\text{ns}$

Processing delay =  $0.5\text{ns} \times 100 \text{ bytes} + 3000\text{ns} = 3050\text{ns}$

Total delay =  $28150\text{ns}$

Percentage that is propagation delay =  $25000\text{ns}/28150\text{ns} = 0.888099 = 88.8099\%$

WAN:

Propagation delay =  $5000\text{km} / 200000\text{km/s} = 25000000\text{ns}$

Transmission delay =  $100\text{ns}$

Processing delay =  $0.5\text{ns} \times 100 \text{ bytes} + 30000\text{ns} = 30050\text{ns}$

Total delay =  $25030150\text{ns}$

Percentage that is propagation delay =  $25000000\text{ns}/25030150\text{ns} = 0.998795 = 99.8795\%$

### Question 2

a.

Transmission delay =  $3 \times 4000 \text{ bits} / 100000000 \text{ bits/s} = 0.12\text{ms}$

Propagation delay =  $2 \times 1\text{ms} = 2\text{ms}$

Total delay =  $2.12\text{ms}$

b.

Total delay =  $(k - 1) \times \text{transmission delay} + n \times \text{transmission delay}$  where transmission delay per packet is equal to  $(H + M/n)/B$

$$f(n) = (k + n - 1) \frac{(H + M/n)}{B}$$

$$f'(n) = \frac{H}{B} - \frac{(k-1)M}{Bn^2}$$

$$0 = \frac{H}{B} - \frac{(k-1)M}{Bn^2}$$

$$\frac{(k-1)M}{Bn^2} = \frac{H}{B}$$

$$\frac{(k-1)M}{H} = n^2$$

$$n = \sqrt{\frac{(k-1)M}{H}}$$

$$f''(n) = \frac{2(k-1)M}{Bn^3}$$

$F''(n)$  is always positive because  $k$  must be greater than or equal to two. This means that the value of  $n$  in which  $F'(n)$  is zero is the global minimum.

### Question 3

a.

Nyquists:

$$\text{Symbols/sec} = 2H = 2 * 400\text{kHz} = 800\text{kbps}$$

$$\text{Bps} = 800\text{kHz} * \log_2(2) = 800\text{kbps}$$

Shannons:

$$\text{Bps} = 400\text{kHz} * \log_2(64) = 2.4\text{Mbps}$$

The tighter upper bound is Nyquist's with it being 800kbps.

b.

Nyquists:

$$\text{Symbols/sec} = 2H = 2 * 400\text{kHz} = 800\text{kbps}$$

$$\text{Bps} = 800\text{kHz} * \log_2(32) = 4\text{Mbps}$$

Shannons:

$$\text{Bps} = 400\text{kHz} * \log_2(64) = 2.4\text{Mbps}$$

The tighter upper bound is Shannon's with it being 2.4Mbps.

**Question 4**

a.

$$10 \text{ Mbps} / 100 \text{ users} = 100 \text{ Kbps} / \text{user}$$

b.

$$B = 10 \text{ seconds}$$

$$\text{Average number of users} = 99 * (10/11) + 1 = 91$$

$$\text{Average achievable data rate} = 10 \text{ Mbps} / 91 = 109.89 \text{ kbps}$$

$$B = 1 \text{ second}$$

$$\text{Average number of users} = 99 * (1/2) + 1 = 50.5$$

$$\text{Average achievable data rate} = 10 \text{ Mbps} / 50.5 = 198.02 \text{ kbps}$$

$$B = 100 \text{ milliseconds}$$

$$\text{Average number of users} = 99 * (0.1/1.1) + 1 = 10$$

$$\text{Average achievable data rate} = 10 \text{ Mbps} / 10 = 1 \text{ Mbps}$$

$$B = 10 \text{ milliseconds}$$

$$\text{Average number of users} = 99 * (0.01/1.01) + 1 = 1.98$$

$$\text{Average achievable data rate} = 10 \text{ Mbps} / 1.98 = 5.05 \text{ Mbps}$$